Supplementary file 9. Risk of bias assessment in included studies

Risk of bias assessment of randomized controlled trials
(Please indicate whether low, some concerns, and high)

Author	Bias arising from the randomization process	Bias due to deviations from intended intervention	Bias due to missing outcome data	Bias in measurement of the outcome	Bias in selection of the reported result	Overall risk of bias
Elbourne 1987	Low	High	High	Some concerns	High	High
Gholipour 2018	Low	Low	Low	Some concerns	Some concerns	High
Grøvdal, 2006	Low	Some concerns	Low	High	Low	High
Osaki 2018	Some concerns	Some concerns	Low	High	Low	High

Risk of bias assessment of quasi-experimental studies
(Please indicate whether low, moderate, serious, critical, no information)

Author	Selection of participants	Confounding variables	Classification of interventions	Deviations from intended interventions	Missing data	Measurement of the outcome	Selection of the reported result	Overall risk of bias
Hagiwara 2013	Low	Moderate	Low	Moderate	Low	Low	Low	Moderate
Jeffs 1994	Low	Moderate	Low	Moderate	Low	Low	Low	Moderate
Moore 2000	Low	Moderate	Low	Serious	Moderate	Moderate	Moderate	Serious
Shah 1993	Low	Serious	Moderate	Moderate	Serious	Serious	Serious	Serious

Risk of bias assessment of observational cohort and cross-sectional studies
(Please indicate whether yes, no, CD [cannot determine], NA [not applicable], NR [not reported])

Author	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Overall
Aihara 2006	Yes	Yes	CD	Yes	No	Yes	Yes	Yes	Yes	No	Yes	NR	NA	Yes	Good
Akiba 2016	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	No	Yes	NR	NA	No	Fair
Aoki 2009	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	Yes	NA	NA	No	Fair
Du Plessis 2017	Yes	Yes	CD	Yes	No	Yes	Yes	Yes	Yes	No	Yes	NR	NA	No	Fair
Fujimoto 2001	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Yes	No	No	No	Fair
Hampshire 2004	Yes	Yes	CD	Yes	No	Yes	Yes	Yes	Yes	No	Yes	NR	NA	Yes	Good
Harrison 1998	Yes	Yes	CD	Yes	No	Yes	Yes	No	No	No	Yes	NR	NA	NA	Fair
Hokama 2000	Yes	Yes	CD	Yes	No	No	Yes	No	Yes	No	Yes	NA	NA	Yes	Fair
Ikeda 2020	Yes	Yes	Yes	Yes	No	No	Yes	No	Yes	No	Yes	NA	NA	NR	Good
Matsumoto 1996	Yes	Yes	Yes	Yes	NR	Yes	Yes	Yes	Yes	No	Yes	NA	NA	No	Good
McMaster 1996	Yes	Yes	CD	No	No	Yes	No	No	Yes	No	Yes	NR	NA	No	Fair
Naito 2019	Yes	Yes	NA	Yes	No	No	Yes	Yes	Yes	No	Yes	NA	NA	Yes	Fair
O'Flaherty 1987	No	Yes	Yes	Yes	NR	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Fair
Ogasawara 2016	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No	Yes	No	NA	No	Fair
Ogawa 2021	Yes	Yes	Yes	Yes	No	No	Yes	No	Yes	No	Yes	NA	NA	NR	Good
Polnay 1989	No	Yes	Yes	Yes	NR	Yes	Yes	Yes	Yes	Yes	Yes	NR	NR	No	Good
Shimizu 2007	Yes	CD	CD	Yes	No	No	Yes	No	No	No	Yes	NA	NR	NA	Poor
Sugi 1985	Yes	Yes	CD	Yes	No	No	Yes	Yes	Yes	No	Yes	No	NA	NR	Fair
Takeda 2002	Yes	Yes	NR	Yes	No	No	No	Yes	Yes	No	Yes	NA	NA	Yes	Fair
Tanabe 2011	Yes	Yes	NR	Yes	No	Yes	Yes	Yes	Yes	No	Yes	NA	NA	Yes	Good
Umeda 2015	Yes	Yes	NR	Yes	No	No	No	No	No	No	Yes	NA	NA	No	Poor
Walton 2007	Yes	Yes	CD	No	No	Yes	NR	NA	Yes	No	Yes	No	NA	No	Fair
Wright 2005	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	NA	No	No	Good
Yuge 2010	Yes	Yes	No	Yes	No	No	Yes	Yes	Yes	No	Yes	NR	NA	No	Fair

1: Was the research question or objective in this paper clearly stated? **2:** Was the study population clearly specified and defined? **3:** Was the participation rate of eligible persons at least 50%? **4:** Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study prespecified and applied uniformly to all participants? **5:** Was a sample size justification, power description, or variance and effect estimates provided? **6:** For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured? **7:** Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed? **8:** For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure, or exposure measured as continuous variable)? **9:** Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants? **10:** Was the exposure(s) assessed more than once over time? **11:** Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants? **12:** Were the outcome assessors blinded to the exposure status of participants? **13:** Was loss to follow-up after baseline 20% or less? **14:** Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?

Risk of bias assessment of qualitative studies
(Please indicate whether yes, no, or can't tell)

Author	1	2	3	4	5	6	7	8	9	10	Overall
Clendon 2010	Yes	Yes	Yes	Can't tell	Yes	Can't tell	Yes	Yes	Yes	Yes	Good
Engida 2013	Yes	Yes	Yes	Yes	Yes	Can't tell	Can't tell	No	Yes	Yes	Good
Fujii 2020	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes	Good
Higashiyama 2013	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes	Good
Lee 2016	Yes	Yes	Yes	Yes	Yes	Can't tell	No	Yes	Yes	Yes	Good
McKinn 2017	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Good
Minewaki 2019	Yes	Yes	Yes	Can't tell	Can't tell	No	Yes	Yes	Yes	Yes	Good
Phipps 2001	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	No	Yes	Yes	Good
Seto 2006	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Yes	Yes	No	Yes	Good
Whitford 2014	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes	Good
Yahata 2005	Yes	No	No	No	No	No	Yes	No	Yes	Yes	Fair
Young 1990	Yes	No	Yes	Yes	Yes	Can't tell	Can't tell	No	Yes	Yes	Good

1: Was there a clear statement of the aims of the research? **2:** Is a qualitative methodology appropriate? **3:** Was the research design appropriate to address the aims of the research? **4:** Was the recruitment strategy appropriate to the aims of the research? **5:** Was the data collected in a way that addressed the research issue? **6:** Has the relationship between researcher and participants been adequately considered? **7:** Have ethical issues been taken into consideration? **8:** Was the data analysis sufficiently rigorous? **9:** Is there a clear statement of findings? **10:** Is the research valuable?

Risk of bias assessment of mixed methods studies
(Please indicate whether yes, no, or can't tell)

Author	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Overall
Bhuiyan 2006	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Good
Grippo 2007	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Good
Hamilton 2012	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Good

1: Is there an adequate rationale for using a mixed methods design to address the research question? **2:** Are the different components of the study effectively integrated to answer the research question? **3:** Are the outputs of the integration of qualitative and quantitative components adequately interpreted? **4:** Are divergences and inconsistencies between quantitative and qualitative results adequately addressed? **5:** Do the different components of the study adhere to the quality criteria of each tradition of the methods involved? **6:** Is the qualitative approach appropriate to answer the research question? **7:** Are the qualitative data collection methods adequate to address the research question? **8:** Are the findings adequately derived from the data? **9:** Is the interpretation of results sufficiently substantiated by data? **10:** Is there coherence between qualitative data sources, collection, analysis and interpretation? Questions 11-15 depends on whether it involves RCT, non-randomized, or quantitative descriptive studies.